

# Supplementary Material:

## Spatiotemporal Aliasing in Video Action Recognition:

### A Cross-Architecture Analysis at Scale

Anonymous ACCV 2026 Submission

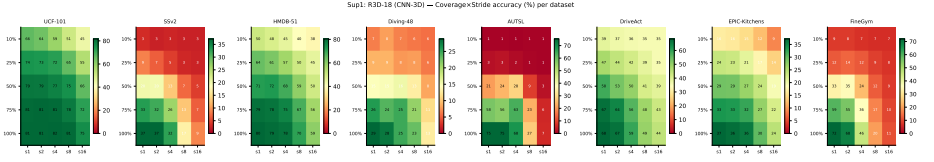
Paper ID #\*\*\*\*\*

This document provides supplementary material for the main paper. We include: (S1) full coverage $\times$ stride heatmaps for all 8 models and 8 datasets; (S2) Levene variance inflation analysis; (S3) full ANOVA effect size tables including FineGym; (S4) action sensitivity taxonomy detail; (S5) entropy routing curves for all models; (S6) clip duration analysis; (S7) spectral correlation detail; (S8) implementation details (target-resolution fine-tuning, positional embedding interpolation); (S9) full multi-dataset spatial resolution results (all 8 datasets  $\times$  8 models); (S10) interactive dashboard and code availability; (S11) TDS ranking robustness: leave-one-architecture-out, architecture-family ablation (CNN-only, Transformer-only, Transformer+SSM pools), and bootstrap confidence intervals; (S12) a direct spectral test of the Nyquist framing across 8 datasets  $\times$  5 resolutions; and (S13) additional robustness checks on optimization headroom, coverage-stride interaction, checkpoint comparability, and entropy-threshold calibration.

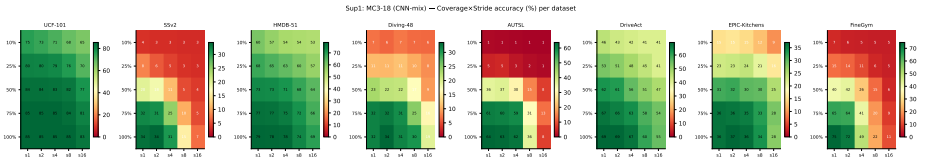
## S1 Full Coverage $\times$ Stride Heatmaps

Figures S1–S8 show the complete coverage $\times$ stride accuracy heatmaps for all 8 architectures across all 8 datasets. Each cell represents Top-1 accuracy (%) at a specific (coverage, stride) configuration. Color scale: green=high accuracy, red=low accuracy. FineGym heatmaps (all 8 models) are included; per-class optical-flow analysis for FineGym is reported in supplementary S7.

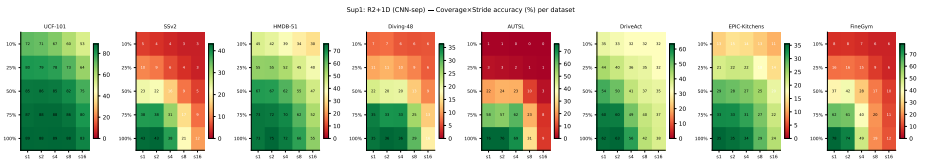
Row order (coverage): 10%, 25%, 50%, 75%, 100%. Column order (stride): 1, 2, 4, 8, 16.



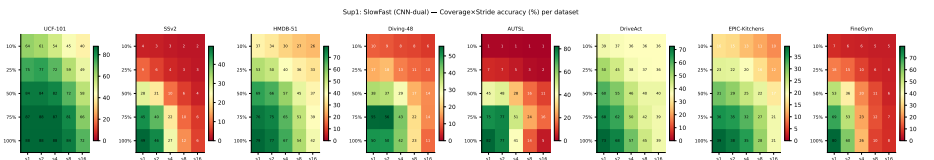
**Fig. S1:** R3D-18 (CNN-3D) coverage×stride accuracy (%) across 8 datasets. Sharp cliff at stride 4→8 on AUTSL (75%→27%) and similar cliff on FineGym confirms the high temporal demand of fine-grained actions. EPIC-Kitchens and UCF-101 retain high accuracy even at sparse sampling.



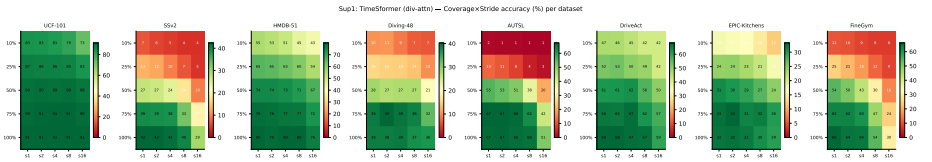
**Fig. S2:** MC3-18 (CNN-mix, channel-separated 3D) heatmaps. Similar pattern to R3D-18 but with slightly reduced aliasing on low-TDS datasets (UCF-101, DriveAct), consistent with the regularizing effect of 2D-channel separation. FineGym shows the same severe cliff pattern as AUTSL.



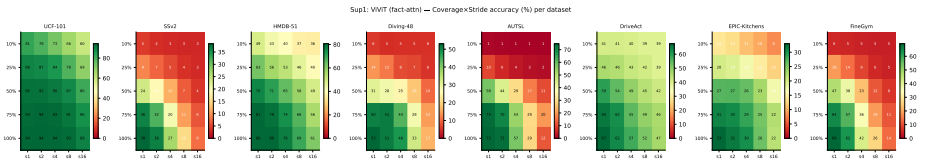
**Fig. S3:** R2+1D-18 (separable spatiotemporal convolution) heatmaps. Higher base accuracy than R3D-18 on several datasets (SSv2: 42.6% vs 37.1%), but similar aliasing sensitivity pattern—cliff at stride 4→8 on AUTSL and FineGym.



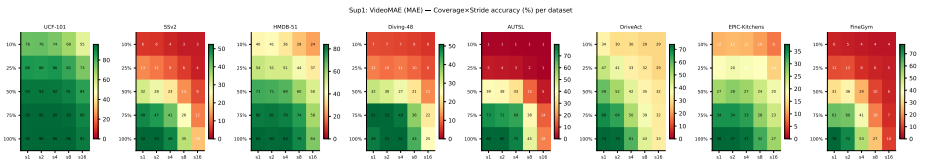
**Fig. S4:** SlowFast-R50 (dual-pathway) heatmaps. Cliff occurs at stride 2→4 on AUTSL (77%→41%)—one step earlier than other CNNs—because the fast pathway requires 32 frames. On FineGym, SlowFast achieves the highest base accuracy (78.2%) among all models but suffers the worst aliasing loss (71.5pp at stride=16).



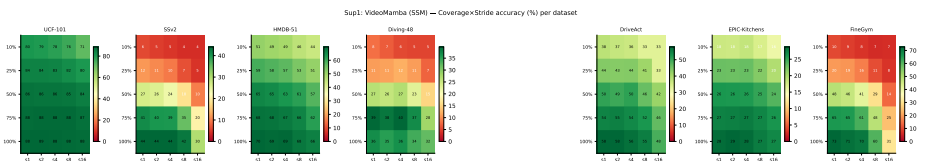
**Fig. S5:** TimeSformer (divided space-time attention) heatmaps. Notably flat across most stride values, with steep degradation only at the combined extreme of low coverage *and* high stride. FineGym exception: drops 36.1pp at stride=16, revealing that even global divided attention cannot fully compensate for the 6–8Hz gymnastics phase transitions.



**Fig. S6:** ViViT (factorized attention) heatmaps. Despite being a Transformer, ViViT aliases nearly as severely as 3D CNNs (AUTSL: cliff at stride 2→4, identical to Slow-Fast). FineGym shows a 55.2pp drop at stride=16, confirming that factorized temporal attention leaves the model vulnerable to frame dropout in high-TDS domains.



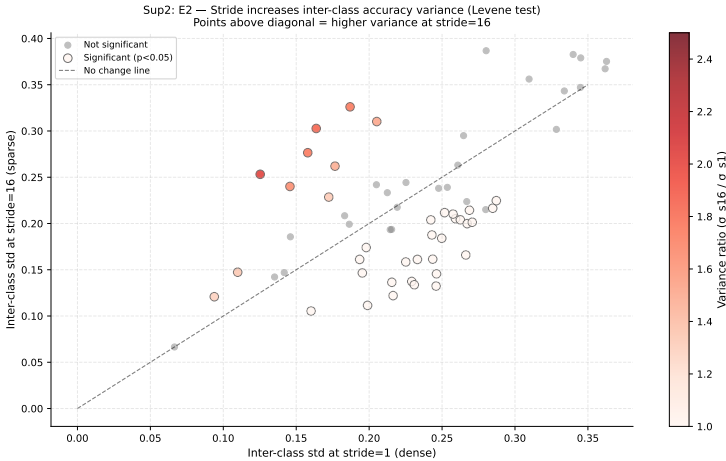
**Fig. S7:** VideoMAE (masked autoencoder) heatmaps. High base accuracy (SSv2: 52.4%, UCF-101: 95.4%, FineGym: 78.0%) but steep temporal degradation (AUTSL cliff at stride 4→8; FineGym: 67.8pp drop). The MAE pretraining objective is spatial—reconstructing patches—and does not transfer to temporal robustness.



**Fig. S8:** VideoMamba (SSM selective scan) heatmaps (all 8 datasets). On 5 of 8 datasets, VideoMamba shows a flat heatmap with minimal degradation across strides—the SSM recurrent state aggregates temporal context, providing natural robustness. FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp) are exceptions, revealing the limit of SSM temporal integration under high-frequency transitions.

## S2 Levene Variance Inflation Analysis

Figure S9 plots inter-class standard deviation at stride=1 versus stride=16 for all model-dataset pairs. Points above the diagonal indicate that sparse sampling increases inter-class variability. Significant pairs ( $p < 0.05$ , Bonferroni-corrected) are highlighted in red.



**Fig. S9:** Levene test: inter-class accuracy std at stride=1 vs. stride=16. Points above the dashed diagonal have higher variance under sparse sampling. Color encodes the variance ratio ( $\sigma_{s16}/\sigma_{s1}$ ). **Key finding:** 67% of model-dataset pairs show significant variance inflation ( $p < 0.05$ ), with the largest increases in VideoMAE/HMDB-51 ( $2.0\times$ ) and R3D-18/HMDB-51 ( $1.85\times$ ). TimeSformer and VideoMamba show stable variance ( $\approx 1.1\times$ ).

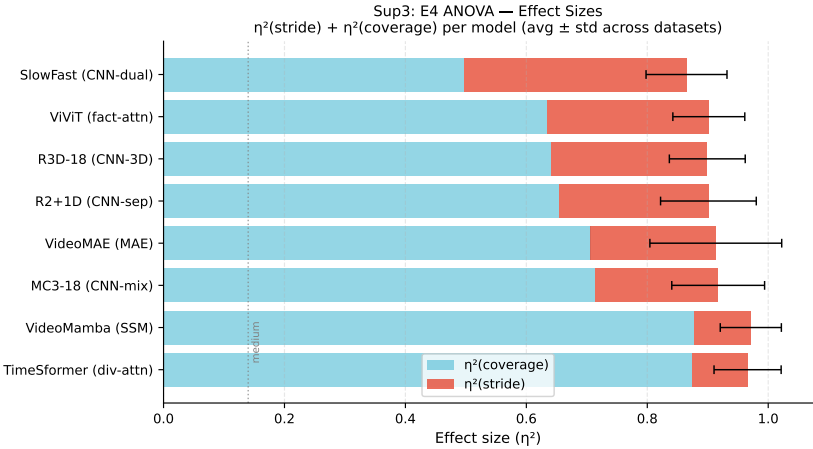
Table S1 lists the 10 model-dataset pairs with highest variance inflation, quantifying the instability masked by mean accuracy.

**Table S1:** Top-10 variance inflation cases (stride=1→16, coverage=100%). All rows significant at  $p<0.05$ .

| Model    | Dataset | $\sigma_{s=1}$ | $\sigma_{s=16}$ | Ratio        | Levene $p$ |
|----------|---------|----------------|-----------------|--------------|------------|
| VideoMAE | HMDB-51 | 0.125          | 0.253           | $2.02\times$ | $<0.0001$  |
| R3D-18   | HMDB-51 | 0.164          | 0.303           | $1.85\times$ | 0.0001     |
| ViViT    | HMDB-51 | 0.158          | 0.277           | $1.75\times$ | 0.0004     |
| SlowFast | HMDB-51 | 0.187          | 0.326           | $1.74\times$ | $<0.0001$  |
| SlowFast | UCF-101 | 0.146          | 0.240           | $1.65\times$ | $<0.0001$  |
| R2+1D    | HMDB-51 | 0.205          | 0.310           | $1.51\times$ | 0.0021     |
| MC3-18   | HMDB-51 | 0.177          | 0.262           | $1.48\times$ | 0.0187     |
| ViViT    | UCF-101 | 0.110          | 0.147           | $1.34\times$ | 0.0093     |
| VideoMAE | AUTSL   | 0.172          | 0.229           | $1.33\times$ | 0.0324     |
| VideoMAE | UCF-101 | 0.094          | 0.121           | $1.29\times$ | 0.0239     |

S3 Full ANOVA Effect Size Tables

Figure S10 shows the mean  $\eta^2$  per model (averaged over all valid datasets), with error bars indicating standard deviation.



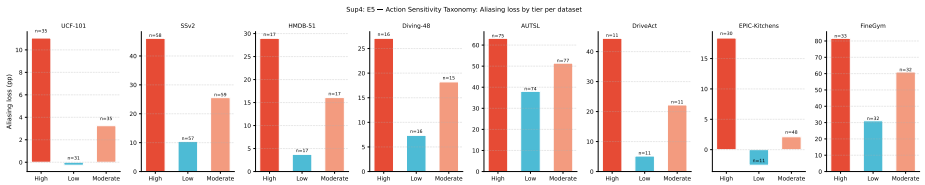
**Fig. S10:** ANOVA effect sizes:  $\eta^2(\text{coverage}) + \eta^2(\text{stride})$  per model. Coverage consistently dominates (blue bars, 53–90% of variance). The stride effect is large for CNNs and ViViT (red bars, 18–35%) but small for VideoMamba and TimeSformer ( $<12\%$ ), confirming architecture-dependent aliasing. Error bars: std across all 8 datasets.

Table S2 provides the complete per-dataset  $\eta^2$  values for all 8 datasets. All stride effects are significant ( $p < 0.05$ ); coverage effects are significant ( $p < 0.001$ ) in all cases. FineGym ( $\eta^2_{\text{stride}}$  up to 0.50 for SlowFast) and AUTSL show the largest stride effects of all 8 datasets, consistent with their co-equal top ranking on TDS (FineGym=55.9pp, AUTSL=53.0pp; see main paper Sec. 3.1 and supplementary S11 for the leave-one-out sensitivity of their relative order).

**Table S2:** Full ANOVA  $\eta^2$  per model and dataset. Cov =  $\eta^2$ (coverage), Str =  $\eta^2$ (stride). Rows shaded darker indicate larger stride effect. FineGym column added with values computed from the full coverage×stride sweep.

| Model  | AUTSL |     | FineGym |     | SSv2 |     | Diving-48 |     | HMDB-51 |     | DriveAct |     | EPIC |     | UCF-101 |     |
|--------|-------|-----|---------|-----|------|-----|-----------|-----|---------|-----|----------|-----|------|-----|---------|-----|
|        | Cov   | Str | Cov     | Str | Cov  | Str | Cov       | Str | Cov     | Str | Cov      | Str | Cov  | Str | Cov     | Str |
| VMamba | .88   | .09 | .77     | .16 | .88  | .09 | .94       | .04 | .90     | .09 | .83      | .15 | .95  | .05 | .88     | .07 |
| TSF    | .92   | .06 | .74     | .19 | .87  | .10 | .97       | .02 | .92     | .07 | .84      | .14 | .91  | .08 | .85     | .07 |
| MC3    | .66   | .18 | .50     | .33 | .54  | .28 | .81       | .13 | .81     | .18 | .75      | .22 | .89  | .11 | .79     | .18 |
| R3D    | .60   | .21 | .47     | .34 | .60  | .26 | .77       | .15 | .68     | .31 | .58      | .33 | .75  | .24 | .70     | .25 |
| R2+1D  | .63   | .20 | .50     | .32 | .62  | .26 | .76       | .17 | .72     | .27 | .49      | .38 | .80  | .18 | .76     | .20 |
| VMAE   | .69   | .18 | .53     | .28 | .69  | .22 | .82       | .12 | .84     | .16 | .43      | .43 | .87  | .11 | .81     | .14 |
| ViViT  | .61   | .24 | .53     | .32 | .51  | .33 | .69       | .20 | .71     | .27 | .60      | .31 | .80  | .19 | .68     | .26 |
| SF     | .51   | .29 | .26     | .50 | .44  | .36 | .51       | .33 | .55     | .39 | .39      | .46 | .67  | .30 | .61     | .36 |

S4 Action Sensitivity Taxonomy



**Fig. S11:** Aliasing loss (pp) per sensitivity tier and dataset. Each bar shows mean accuracy drop (stride=1→16) within the tier, annotated with the number of action classes. UCF-101 Low tier: −0.3pp (static, appearance-dominated actions lose essentially no accuracy under sparse sampling). AUTSL: all three tiers remain high (> 38pp), confirming that sign language is fundamentally high-frequency regardless of action subclass.

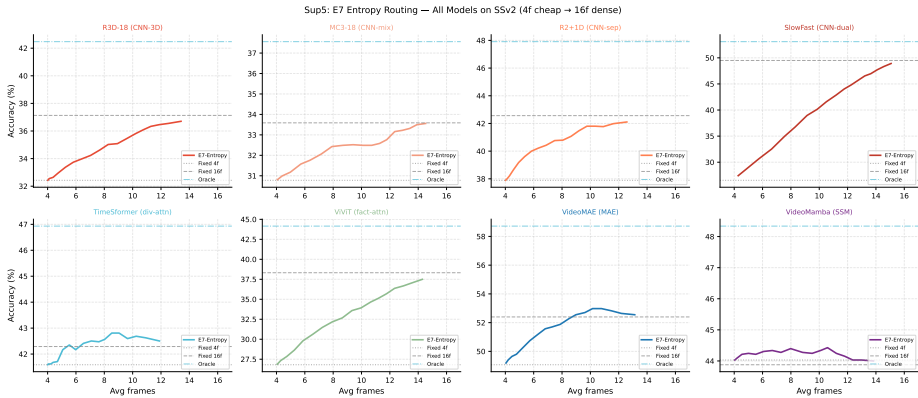
Table S3 lists the threshold values that define tier boundaries per dataset, derived from the 33rd and 67th percentiles of the per-class accuracy drop.

**Table S3:** Taxonomy tier thresholds (accuracy drop, pp): classes below the Low threshold alias negligibly; classes above the High threshold are severely affected. Thresholds are dataset-specific and computed across all valid architectures.

| Dataset       | Classes | Low threshold | High threshold | High count | Low count |
|---------------|---------|---------------|----------------|------------|-----------|
| AUTSL         | 226     | 66.9pp        | 73.9pp         | 75         | 74        |
| Diving-48     | 47      | 25.0pp        | 47.7pp         | 16         | 16        |
| SSv2          | 174     | 59.7pp        | 76.9pp         | 58         | 57        |
| HMDB-51       | 51      | 8.4pp         | 28.5pp         | 17         | 17        |
| EPIC-Kitchens | 89      | 0.0pp         | 19.4pp         | 30         | 11        |
| DriveAct      | 33      | 14.8pp        | 51.0pp         | 11         | 11        |
| UCF-101       | 101     | 1.4pp         | 7.5pp          | 35         | 31        |

S5 Entropy Routing Curves — All Models

Figure S12 shows accuracy vs. average frames for all 8 architectures on SSv2 under the entropy routing protocol. Each curve is the result of sweeping confidence threshold  $\tau \in [0.10, 0.95]$ . Dashed horizontal lines indicate fixed-budget baselines (4f, 16f) and the oracle.



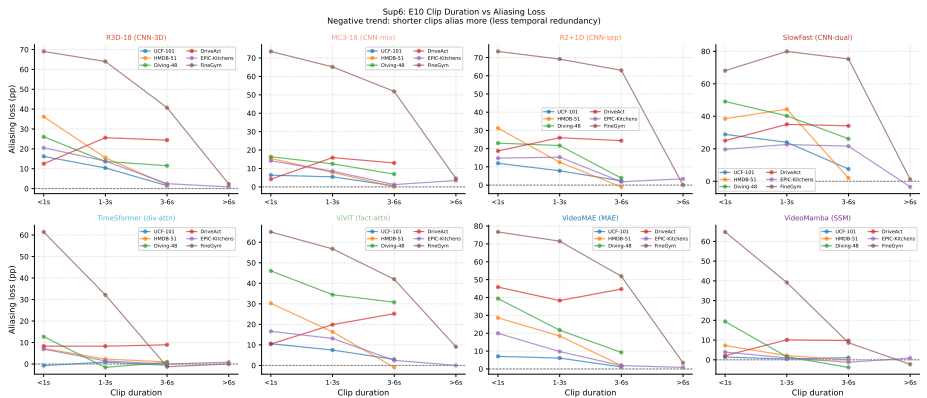
**Fig. S12:** Entropy routing: accuracy vs. avg frames (SSv2, 4f cheap → 16f dense). For architectures with flat 4f≈16f accuracy (TimeSformer, VideoMamba), routing near-trivially selects 4f for most videos and matches dense accuracy. For architectures with large 4f–16f gaps (SlowFast, ViViT), the routing curve provides meaningful operating points between the fixed-budget extremes.

Table S4 provides the full entropy routing results across all models and datasets at the optimal threshold (avg frames  $\leq 8$ ).

**Table S4:** Entropy routing: best accuracy at avg  $\leq 8$  frames per model/dataset. Values are Top-1 accuracy (%). Bold = highest per column. \*VideoMamba/AUTSL: entropy routing not computed; value is direct stride=8 accuracy (65.0%).

| Model       | AUTSL       | FineGym     | Diving      | SSv2        | HMDB        | DriveAct    | EPIC        | UCF         | Avg         |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| R3D-18      | 73.3        | 68.5        | 27.6        | 34.6        | 80.2        | 66.1        | 36.7        | 81.6        | 58.6        |
| MC3-18      | 63.4        | 62.9        | 31.8        | 32.4        | 78.5        | 69.4        | 36.5        | 85.6        | 57.6        |
| R2+1D       | 74.2        | 72.9        | 36.3        | 40.8        | 73.3        | 59.6        | 34.8        | 89.0        | 60.1        |
| SlowFast    | 59.5        | 57.8        | 46.6        | 35.0        | 76.3        | 67.4        | 37.4        | 87.9        | 58.5        |
| TimeSformer | 67.1        | 66.8        | 38.9        | 42.5        | 80.6        | 68.5        | 32.4        | 91.3        | <b>61.0</b> |
| ViViT       | 67.2        | 66.1        | 48.9        | 32.2        | 80.1        | 64.3        | 31.3        | 94.3        | 60.6        |
| VideoMAE    | <b>79.5</b> | <b>76.4</b> | <b>52.6</b> | <b>51.9</b> | <b>84.7</b> | <b>69.0</b> | <b>37.8</b> | <b>95.9</b> | <b>68.5</b> |
| VideoMamba  | 65.0*       | 73.2        | 37.4        | 44.4        | 69.8        | 57.6        | 28.4        | 88.5        | 58.0        |

S6 Clip Duration Analysis



**Fig. S13:** Aliasing loss (pp) vs. clip duration bin for all 8 architectures across multiple datasets. Counter-intuitive finding: **shorter clips alias more** (Pearson  $r = -0.3$  to  $-0.8$  across models). Short clips have less temporal redundancy; any frame drop removes a larger fraction of the available evidence. Long clips ( $>4s$ ) contain natural temporal repetition that compensates for sparse sampling. Exception: Diving-48 longer clips ( $>4s$ ) alias more, consistent with longer dives containing more distinct phase transitions.

S7 Spectral Correlation Detail

We estimated per-class temporal frequency demand using optical-flow magnitude (Farneback algorithm [1]) computed on 5 representative videos per class, sampling 16 frames per video at  $112 \times 112$ px. The mean inter-frame flow magnitude serves as a proxy for temporal activity level. This analysis is exploratory



and should not be read as confirmatory evidence for the Nyquist-aliasing hypothesis on its own; see the main paper (Sec. 3.1, Fig. 3 right) for the summary figure including FineGym.

Table S5 reports Pearson correlation between per-class mean flow magnitude and aliasing loss (accuracy drop stride=1→16) per dataset. Prior studies on small closed-vocabulary datasets (UCF-101, Kinetics-400) have reported near-perfect optical-flow-aliasing correlations ( $r=0.99$ ) using controlled synthetic flow protocols. Our large-scale evaluation on 8 diverse datasets yields a more nuanced picture ( $r = -0.475$  to  $0.327$ ). This has at least four plausible explanations: (1) complex background motion in egocentric datasets (EPIC-Kitchens) decorrelates flow from action content; (2) sign language (AUTSL) involves localized, subtle movements that optical flow at 112px partially misses; (3) our use of raw flow magnitude rather than class-conditioned frequency analysis is a coarser proxy; (4) FineGym’s negative correlation suggests that for discrete-event classes (a held position, a precise landing), raw motion magnitude is the wrong signal entirely—what matters is whether sparse sampling catches the one brief moment that defines the class, which low-flow (slow, precise) actions are more likely to lose than high-flow (fast, sustained) ones.

**Table S5:** Spectral correlation: Pearson  $r$  between optical-flow magnitude and per-class aliasing loss. Positive  $r$  supports the Nyquist hypothesis; near-zero or negative values suggest flow magnitude is an insufficient (or inverted) proxy for temporal demand.

| Dataset       | Pearson $r$ | $p$ -value | Significant ( $p<0.05$ ) |
|---------------|-------------|------------|--------------------------|
| DriveAct      | 0.327       | 0.065      | Marginal                 |
| AUTSL         | 0.285       | <0.001     | Yes                      |
| SSv2          | 0.184       | <0.001     | Yes                      |
| HMDB-51       | 0.180       | 0.206      | No                       |
| Diving-48     | 0.140       | 0.349      | No                       |
| UCF-101       | 0.031       | 0.762      | No                       |
| EPIC-Kitchens | -0.002      | 0.985      | No                       |
| FineGym       | -0.475      | <0.0001    | Yes (inverted)           |

## S8 Implementation Details

*Hardware.* Training experiments used NVIDIA A100 (40GB) and H200 NVL (141GB) GPUs (CUDA 13.2). Temporal-sweep evaluation (1,600 configurations) was parallelized across 8 nodes using a custom Slurm auto-submitter. Inference latency measurements used an NVIDIA RTX PRO 6000 Blackwell GPU (96GB) with bfloat16 autocast, batch size=1, 20 warmup iterations, 100 benchmark iterations, and CUDA event timing with explicit synchronization.

*Training hyperparameters.* All models fine-tuned for 10 epochs from Kinetics-400 checkpoints. Transformers and VideoMamba: AdamW, lr= $2 \times 10^{-5}$ , weight

decay=0.05, batch size=32 per GPU. CNNs: AdamW, lr= $10^{-4}$ , batch size=128 per GPU. SlowFast: AdamW, lr= $10^{-4}$ , batch size=32 per GPU. All models use cosine learning rate decay with linear warmup (1 epoch). Spatial augmentation: random resized crop, horizontal flip, normalization (ImageNet statistics).

*Positional embedding interpolation for Transformers and SSMs.* For target-resolution fine-tuning at non-native resolutions, Transformer and SSM positional embeddings must be adapted. *The wrong approach:* using the HuggingFace mismatched-size flag discards positional embeddings entirely, causing a large accuracy cliff (typically 10–20pp worse than our approach). *Our approach:* load the native-resolution checkpoint, then bicubically interpolate spatial positional embeddings from the native grid ( $14 \times 14 = 196$  tokens at 224px) to the target grid ( $\lfloor r/16 \rfloor^2$  tokens at resolution  $r$ ) before copying weights to a model instantiated at the target resolution. This preserves the spatial topology learned during pre-training and ensures fine-tuning starts from a meaningful positional encoding rather than a random one. The same bicubic interpolation applies to TimeSformer, VideoMAE, ViViT, and VideoMamba.

*SlowFast spatial pooling fix.* The original SlowFast-R50 head uses a fixed  $7 \times 7$  average pooling kernel, assuming 224px input. For non-native resolutions, we replace both head pooling layers with `AdaptiveAvgPool3d(1)`, enabling forward passes at arbitrary input resolutions without parameter changes.

*FineGym dataset details.* FineGym [2] contains 97 fine-grained gymnastics action classes (floor exercise, uneven bars, balance beam, vault; parallel bars, horizontal bar, pommel horse, rings for men). We use the FineGym288 subset (1,960 clips), balanced to 20 clips per class for training and evaluation. Preprocessing: clips are trimmed to the annotated action boundary (mean duration 3.2s at 25fps) and padded with the last frame if shorter than 16 frames.

## S9 Multi-Dataset Spatial Resolution Results

Tables S6–S13 extend the spatial robustness analysis from the main paper (Section 4.3) to all 8 datasets. Tables S6–S12 show cross-resolution results (no re-training): the fine-tuned native-resolution checkpoint evaluated at 96–336px. Table S13 shows FineGym target-resolution fine-tuning results (one fine-tuned checkpoint per resolution). **Bold** = native training resolution.  $\Delta_{\max}$  = largest accuracy drop from native across all tested resolutions.

**Table S6:** Spatial robustness on **SSv2** (repeated from main paper; no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$ |
|-------------|--------|------|-------------|-------|-------------|-------|-----------------|
| R3D-18      | 112px  | 35.9 | <b>37.1</b> | 30.3  | 17.2        | 5.8   | −31.4pp         |
| MC3-18      | 112px  | 32.4 | <b>33.6</b> | 27.8  | 14.8        | 4.4   | −29.2pp         |
| R2+1D       | 112px  | 40.1 | <b>42.6</b> | 36.2  | 20.8        | 5.9   | −36.6pp         |
| SlowFast    | 224px  | 27.7 | 32.4        | 45.7  | <b>49.5</b> | 36.0  | −21.8pp         |
| TimeSformer | 224px  | 39.3 | 39.4        | 41.4  | <b>42.3</b> | 42.1  | −3.0pp          |
| ViViT       | 224px  | 36.4 | 37.1        | 38.1  | <b>38.3</b> | 38.0  | −1.9pp          |
| VideoMAE    | 224px  | 48.9 | 49.2        | 51.5  | <b>52.3</b> | 51.9  | −3.4pp          |
| VideoMamba  | 224px  | 37.5 | 39.3        | 42.5  | <b>43.9</b> | 43.8  | −6.4pp          |

**Table S7:** Spatial robustness on **UCF-101** (no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$ |
|-------------|--------|------|-------------|-------|-------------|-------|-----------------|
| R3D-18      | 112px  | 79.9 | <b>81.2</b> | 75.3  | 55.1        | 20.7  | −60.5pp         |
| MC3-18      | 112px  | 82.9 | <b>85.4</b> | 81.0  | 63.9        | 24.9  | −60.5pp         |
| R2+1D       | 112px  | 88.0 | <b>88.6</b> | 85.1  | 64.9        | 25.0  | −63.6pp         |
| SlowFast    | 224px  | 69.7 | 75.8        | 87.5  | <b>91.4</b> | 89.2  | −21.7pp         |
| TimeSformer | 224px  | 89.1 | 89.7        | 90.7  | <b>91.5</b> | 90.9  | −2.4pp          |
| ViViT       | 224px  | 92.0 | 93.5        | 94.7  | <b>95.1</b> | 95.0  | −3.1pp          |
| VideoMAE    | 224px  | 95.3 | 95.7        | 96.1  | <b>96.4</b> | 96.3  | −1.1pp          |
| VideoMamba  | 224px  | 76.7 | 78.7        | 82.6  | <b>83.5</b> | 83.8  | −6.7pp          |

**Table S8:** Spatial robustness on **HMDB-51** (no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$ |
|-------------|--------|------|-------------|-------|-------------|-------|-----------------|
| R3D-18      | 112px  | 78.8 | <b>80.3</b> | 75.9  | 59.8        | 24.7  | −55.6pp         |
| MC3-18      | 112px  | 76.7 | <b>78.6</b> | 74.2  | 51.0        | 14.5  | −64.2pp         |
| R2+1D       | 112px  | 74.4 | <b>73.1</b> | 68.4  | 50.5        | 16.2  | −56.9pp         |
| SlowFast    | 224px  | 59.5 | 67.2        | 80.5  | <b>81.6</b> | 79.0  | −22.0pp         |
| TimeSformer | 224px  | 75.9 | 77.2        | 77.5  | <b>78.9</b> | 78.0  | −2.9pp          |
| ViViT       | 224px  | 78.0 | 79.0        | 79.9  | <b>79.5</b> | 80.0  | −1.6pp          |
| VideoMAE    | 224px  | 83.7 | 83.1        | 84.0  | <b>83.8</b> | 84.1  | −0.8pp          |
| VideoMamba  | 224px  | 59.9 | 63.8        | 67.2  | <b>69.8</b> | 68.9  | −9.9pp          |

**Table S9:** Spatial robustness on **Diving-48** (no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$ |
|-------------|--------|------|-------------|-------|-------------|-------|-----------------|
| R3D-18      | 112px  | 26.5 | <b>28.8</b> | 22.1  | 12.8        | 7.3   | −21.5pp         |
| MC3-18      | 112px  | 30.8 | <b>31.6</b> | 27.3  | 13.7        | 7.3   | −24.4pp         |
| R2+1D       | 112px  | 33.8 | <b>35.3</b> | 30.1  | 18.7        | 7.8   | −27.5pp         |
| SlowFast    | 224px  | 18.4 | 25.0        | 43.3  | <b>50.5</b> | 41.6  | −32.0pp         |
| TimeSformer | 224px  | 37.4 | 39.3        | 38.6  | <b>38.1</b> | 38.6  | −0.7pp          |
| ViViT       | 224px  | 48.0 | 49.9        | 52.8  | <b>53.0</b> | 51.3  | −5.0pp          |
| VideoMAE    | 224px  | 48.7 | 48.3        | 50.9  | <b>48.6</b> | 49.1  | −0.3pp          |
| VideoMamba  | 224px  | 29.2 | 29.1        | 36.1  | <b>36.6</b> | 36.2  | −7.5pp          |

**Table S10:** Spatial robustness on **AUTSL** (sign language; no retraining during cross-resolution evaluation). CNN degradation is most severe: R3D-18 collapses from 75.0% to 2.0% at 336px ( $-73.0\text{pp}$ ). VideoMamba is omitted from this cross-resolution table because the available checkpoint is not comparable for this specific spatial sweep.

| Model       | Native | 96px                                        | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$  |
|-------------|--------|---------------------------------------------|-------------|-------|-------------|-------|------------------|
| R3D-18      | 112px  | 72.2                                        | <b>75.0</b> | 63.0  | 27.9        | 2.0   | $-73.0\text{pp}$ |
| MC3-18      | 112px  | 59.1                                        | <b>63.7</b> | 52.7  | 12.2        | 1.2   | $-62.5\text{pp}$ |
| R2+1D       | 112px  | 73.7                                        | <b>75.9</b> | 68.9  | 38.4        | 7.3   | $-68.7\text{pp}$ |
| SlowFast    | 224px  | 22.5                                        | 32.6        | 70.0  | <b>79.2</b> | 69.0  | $-56.7\text{pp}$ |
| TimeSformer | 224px  | 65.1                                        | 66.0        | 68.3  | <b>68.5</b> | 68.6  | $-3.5\text{pp}$  |
| ViViT       | 224px  | 49.2                                        | 51.8        | 53.3  | <b>53.6</b> | 54.0  | $-4.5\text{pp}$  |
| VideoMAE    | 224px  | 77.3                                        | 78.2        | 79.5  | <b>79.6</b> | 79.3  | $-2.2\text{pp}$  |
| VideoMamba  | 224px  | Not included in this cross-resolution sweep |             |       |             |       |                  |

**Table S11:** Spatial robustness on **DriveAct** (in-cabin activities; no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$  |
|-------------|--------|------|-------------|-------|-------------|-------|------------------|
| R3D-18      | 112px  | 65.0 | <b>68.3</b> | 52.7  | 19.0        | 4.9   | $-63.4\text{pp}$ |
| MC3-18      | 112px  | 65.2 | <b>69.0</b> | 38.4  | 6.2         | 4.5   | $-64.5\text{pp}$ |
| R2+1D       | 112px  | 67.4 | <b>62.5</b> | 40.0  | 12.9        | 4.5   | $-58.0\text{pp}$ |
| SlowFast    | 224px  | 31.7 | 40.2        | 65.8  | <b>70.1</b> | 58.7  | $-38.4\text{pp}$ |
| TimeSformer | 224px  | 68.3 | 69.2        | 69.6  | <b>70.3</b> | 69.9  | $-2.0\text{pp}$  |
| ViViT       | 224px  | 59.4 | 58.7        | 62.1  | <b>63.4</b> | 63.6  | $-4.7\text{pp}$  |
| VideoMAE    | 224px  | 71.4 | 69.9        | 71.7  | <b>73.0</b> | 71.4  | $-3.1\text{pp}$  |
| VideoMamba  | 224px  | 35.0 | 38.2        | 46.9  | <b>57.8</b> | 52.9  | $-22.8\text{pp}$ |

**Table S12:** Spatial robustness on **EPIC-Kitchens** (egocentric cooking; no retraining).

| Model       | Native | 96px | 112px       | 160px | 224px       | 336px | $\Delta_{\max}$  |
|-------------|--------|------|-------------|-------|-------------|-------|------------------|
| R3D-18      | 112px  | 33.3 | <b>37.2</b> | 26.7  | 8.6         | 3.0   | $-34.1\text{pp}$ |
| MC3-18      | 112px  | 36.1 | <b>36.2</b> | 27.2  | 9.4         | 2.5   | $-33.6\text{pp}$ |
| R2+1D       | 112px  | 32.8 | <b>35.5</b> | 27.6  | 11.8        | 4.6   | $-30.9\text{pp}$ |
| SlowFast    | 224px  | 11.3 | 17.7        | 31.3  | <b>37.5</b> | 28.7  | $-26.2\text{pp}$ |
| TimeSformer | 224px  | 20.5 | 19.2        | 25.6  | <b>31.7</b> | 23.5  | $-12.5\text{pp}$ |
| ViViT       | 224px  | 30.2 | 34.1        | 36.1  | <b>37.9</b> | 36.0  | $-7.7\text{pp}$  |
| VideoMAE    | 224px  | 30.9 | 32.6        | 36.5  | <b>38.6</b> | 39.1  | $-7.7\text{pp}$  |
| VideoMamba  | 224px  | 11.1 | 12.6        | 20.8  | <b>28.3</b> | 24.5  | $-17.3\text{pp}$ |

**Table S13: FineGym spatial resolution results after target-resolution fine-tuning** (one checkpoint per resolution, 10 fine-tuning epochs from native checkpoint). Bold = native training resolution (112px for CNNs, 224px for Transformers/SSM). Unlike other datasets in S9 (cross-resolution without retraining), FineGym shows CNN performance that *improves beyond native* after fine-tuning at higher resolutions (R3D-18: 71.5%@112px→76.1%@224px), confirming that higher spatial resolution genuinely aids gymnastics pose discrimination.

| Model       | Native | 48px | 96px | 112px       | 160px | 224px       |
|-------------|--------|------|------|-------------|-------|-------------|
| R3D-18      | 112px  | 56.4 | 72.6 | <b>71.5</b> | 74.9  | 76.1        |
| MC3-18      | 112px  | 46.3 | 62.3 | <b>64.3</b> | 73.8  | 65.7        |
| R2+1D       | 112px  | 61.6 | 73.1 | <b>76.8</b> | 79.4  | 76.2        |
| SlowFast    | 224px  | 36.3 | 67.2 | 69.6        | 76.3  | <b>78.2</b> |
| TimeSformer | 224px  | 24.3 | 48.9 | 50.8        | 62.2  | <b>66.4</b> |
| ViViT       | 224px  | 21.3 | 39.1 | 40.8        | 49.6  | <b>69.9</b> |
| VideoMAE    | 224px  | 25.4 | 51.1 | 57.5        | 64.8  | <b>78.0</b> |
| VideoMamba  | 224px  | 29.1 | 51.6 | 58.2        | 66.0  | <b>73.2</b> |

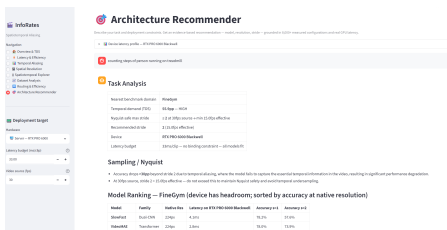
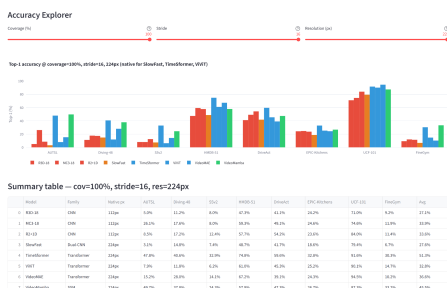
## S10 Interactive Dashboard and Code Availability

*Interactive visualization dashboard.* An anonymized interactive dashboard is included with the supplementary material for review. The public dashboard URL will be released after the review period. The dashboard provides:

1. **Temporal explorer:** coverage×stride heatmaps per model and dataset, with configurable scaling and stride annotations.
2. **Spatial browser:** accuracy curves across resolutions for all 8 models and datasets; target resolution tuning by family.
3. **Recommender:** given a target dataset, latency budget, frame rate, and deployment context, the system combines structured analysis with LLM synthesis (Llama-3.3-70B via Groq) to recommend a model–resolution–stride operating point. Budget-constrained and unconstrained paths are handled separately.
4. **TDS comparison:** rank any subset of the 8 datasets by temporal demand, with model-specific breakdowns and Spearman correlation display.
5. **Routing simulator:** entropy threshold sweep for any model on any dataset, with visual Pareto-frontier display.

*Code availability.* Evaluation scripts, model fine-tuning code, result tables, and the Streamlit dashboard source are provided as an anonymized supplementary package for review. This includes the target-resolution pipeline with bicubic positional embedding interpolation. The public repository will be released after review. The package includes:

- `scripts/accv2026/`: all evaluation and retraining scripts
- `src/info_rates/`: model loading, temporal sweep, entropy routing
- `dashboard/`: Streamlit app with hybrid RAG+LLM recommender
- `evaluations/accv2026/paper_results/`: CSV result files used in the paper.



## S11 Leave-One-Architecture-Out TDS Stability

To assess whether the TDS ranking depends on the specific 8-architecture pool evaluated, we recompute TDS for all 8 datasets after removing each architecture in turn (7-architecture pools) and compare the resulting ranking to the full 8-architecture ranking via Spearman  $\rho$ .

**Table S14:** Leave-one-architecture-out TDS ranking stability.  $\rho$  is the Spearman correlation between the 8-dataset ranking computed with the named architecture excluded (7-architecture pool) and the full 8-architecture ranking.

| Excluded architecture | Spearman $\rho$ | $p$                  |
|-----------------------|-----------------|----------------------|
| R3D-18                | 1.000           | $< 10^{-4}$          |
| MC3-18                | 1.000           | $< 10^{-4}$          |
| R2+1D                 | 1.000           | $< 10^{-4}$          |
| SlowFast              | 1.000           | $< 10^{-4}$          |
| TimeSformer           | 1.000           | $< 10^{-4}$          |
| ViViT                 | 1.000           | $< 10^{-4}$          |
| VideoMAE              | 1.000           | $< 10^{-4}$          |
| VideoMamba            | 0.976           | $3.3 \times 10^{-5}$ |

For 7 of 8 leave-one-out pools, the ranking is *exactly* identical to the full-pool ranking ( $\rho = 1.00$ ). Removing VideoMamba is the single exception ( $\rho = 0.976$ ): it swaps the #1/#2 positions, since excluding VideoMamba raises AUTSL’s average (58.3pp, from 53.0pp) and lowers FineGym’s average (58.2pp, from 55.9pp)—VideoMamba is comparatively robust on AUTSL (16.0pp, below its own median drop) and comparatively fragile on FineGym (40.8pp, its worst dataset), so its presence in the pool is precisely what tips FineGym narrowly ahead of AUTSL. Ranks 3–8 (SSv2 through UCF-101) are unaffected by any single exclusion. We therefore report FineGym and AUTSL as co-equal highest-TDS domains in the main paper rather than asserting a strict order, while the rest of the ranking is robust under leave-one-architecture-out by this test.

### S11.1 Architecture-Family Ablation

Leave-one-out removes a single architecture at a time; a stronger test removes an entire family. We recompute TDS under three restricted pools: **CNN-only** ( $n=4$ : R3D-18, MC3-18, R2+1D, SlowFast), **Transformer-only** ( $n=3$ : TimeSformer, ViViT, VideoMAE), and **Transformer+SSM** ( $n=4$ : adding VideoMamba). Table S15 reports TDS under each pool and Spearman  $\rho$  against the full ranking.

The ranking is essentially invariant to architecture-family composition: removing all four CNNs, keeping only the three Transformers, or keeping Transformers+SSM all leave ranks 3–8 in identical order, and only the CNN-only pool reproduces the same FineGym/AUTSL swap already characterized in the leave-one-out test above ( $\rho = 0.976$  in both cases, for the same reason: CNNs

**Table S15:** TDS per dataset under architecture-family ablation pools, compared to the full 8-architecture ranking (sorted by full-pool TDS).

| Dataset                  | Full ( $n=8$ ) | CNN-only ( $n=4$ ) | Transf.-only ( $n=3$ ) | Transf.+SSM ( $n=4$ ) |
|--------------------------|----------------|--------------------|------------------------|-----------------------|
| FineGym                  | 55.92          | 62.05              | 53.06                  | 49.80                 |
| AUTSL                    | 53.02          | 67.20              | 46.45                  | 38.83                 |
| SSv2                     | 27.29          | 32.16              | 25.95                  | 22.43                 |
| DriveAct                 | 21.48          | 24.05              | 23.21                  | 18.92                 |
| Diving-48                | 19.16          | 21.74              | 20.59                  | 16.59                 |
| HMDB-51                  | 16.47          | 21.47              | 14.54                  | 11.47                 |
| EPIC-Kitchens            | 9.77           | 12.89              | 8.27                   | 6.64                  |
| UCF-101                  | 4.86           | 7.36               | 2.90                   | 2.35                  |
| Spearman $\rho$ vs. full | —              | 0.976              | 1.000                  | 1.000                 |

are comparatively more fragile on AUTSL’s fine handshape transitions than on FineGym’s phase boundaries, which narrows the gap). This directly addresses whether TDS reflects a property of the dataset pool versus an artifact of including any one architecture family: under every family-restricted pool we tested, the ranking is the same dataset ordering reported in the main paper, with the sole exception of the already-disclosed #1/#2 ambiguity.

**S11.2 Bootstrap Confidence Intervals**

To quantify ranking uncertainty directly rather than only testing discrete leave-one-out/family pools, we bootstrap over architectures: we resample 8 architectures with replacement ( $B=10,000$  resamples) from the original pool, recompute TDS for each dataset on each resample, and report the resulting 95% percentile confidence interval in Table S16.

**Table S16:** Bootstrap 95% CI for TDS per dataset (resampling 8 architectures with replacement,  $B=10,000$ ).

| Dataset       | TDS (bootstrap mean) | 95% CI         |
|---------------|----------------------|----------------|
| FineGym       | 55.99                | [47.24, 63.86] |
| AUTSL         | 53.13                | [36.02, 67.03] |
| SSv2          | 27.35                | [20.40, 33.92] |
| DriveAct      | 21.57                | [13.92, 29.35] |
| Diving-48     | 19.23                | [10.67, 28.37] |
| HMDB-51       | 16.54                | [9.53, 24.21]  |
| EPIC-Kitchens | 9.80                 | [6.25, 13.22]  |
| UCF-101       | 4.88                 | [2.18, 8.36]   |

All eight 95% CIs are disjoint from zero, confirming every dataset has a genuine non-zero temporal demand at the architecture-pool level. AUTSL’s CI is the widest (width 31.0pp) because its per-architecture drops are the most heterogeneous (TimeSformer and VideoMamba near-flat, CNNs collapse sharply),



while EPIC-Kitchens and UCF-101 have the narrowest CIs, consistent with their low and relatively architecture-independent TDS.

We additionally bootstrap the FineGym–AUTSL gap directly (same resampling procedure, computing  $\text{TDS}(\text{FineGym}) - \text{TDS}(\text{AUTSL})$  on each resample): mean = +2.92pp, 95% CI =  $[-4.58, +11.73]$ , with FineGym exceeding AUTSL in 73.9% of resamples. The confidence interval includes zero, which is the appropriate statistical test of the co-equal-ranking claim: the data are consistent with FineGym narrowly leading AUTSL in expectation, but do not support rejecting the null hypothesis of equal TDS at the 95% level. This is the quantitative basis, rather than only the leave-one-out swap, for reporting FineGym and AUTSL as co-equal highest-TDS domains in the main paper.

## S12 Direct Spectral Test of the Nyquist Framing

The main paper’s Nyquist–Shannon framing (Sec. ??) is conceptual: we characterize aliasing through accuracy degradation under controlled stride and coverage sweeps rather than by directly measuring action-frequency spectra. Here we report a direct test, intended to give the most literal version of the Nyquist criterion that our evaluation infrastructure supports without retraining any model.

*Methodology.* For each of 8 datasets and 5 resolutions (48, 96, 112, 160, 224 px), we sample up to 12 classes  $\times$  2 videos. We decode up to 64 *consecutive* frames in native order, resize to the target resolution, and compute inter-frame optical-flow magnitude (Farneback) as a temporal signal sampled once per original frame. We estimate each signal’s spectral cutoff frequency  $f_c$  (90% cumulative periodogram power, cycles/frame) and average over sampled videos to obtain one  $f_c$  per (dataset, resolution) pair. We pair each  $f_c$  with empirical stride-sensitivity at the same resolution: mean accuracy drop from stride=1 to stride=16 at coverage=100%, averaged over 8 architectures and read from the existing sweep. This yields  $8 \times 5 = 40$  (dataset, resolution) points.

**Table S17:** Mean flow-derived spectral cutoff frequency  $f_c$ , videos per resolution, and mean empirical stride-sensitivity, averaged over the 5 tested resolutions per dataset and sorted by stride-sensitivity.

| Dataset       | Mean $f_c$ | $n/\text{res.}$ | Mean stride-sens. |
|---------------|------------|-----------------|-------------------|
| FineGym       | 0.188      | 24              | 46.30             |
| AUTSL         | 0.121      | 24              | 45.61             |
| SSv2          | 0.339      | 24              | 25.70             |
| DriveAct      | 0.192      | 21              | 18.65             |
| HMDB-51       | 0.383      | 24              | 16.57             |
| Diving-48     | 0.182      | 24              | 15.21             |
| EPIC-Kitchens | 0.248      | 23              | 9.70              |
| UCF-101       | 0.387      | 24              | 5.45              |

*Result.* Pooling across all 40 (dataset, resolution) points,  $f_c$  is significantly *negatively* correlated with stride-sensitivity (Spearman  $\rho = -0.635$ , permutation  $p = 1.0 \times 10^{-5}$ ,  $n = 40$ )—the opposite sign from the naive Nyquist prediction that higher temporal-frequency content should require denser sampling (and thus show higher measured stride-sensitivity). FineGym integrates cleanly into this pattern: it has the highest mean stride-sensitivity (46.30pp) but a moderate whole-frame cutoff frequency ( $f_c = 0.188$  cycles/frame), close to DriveAct and Diving-48 rather than UCF-101/HMDB-51.

*Interpretation.* This is the same qualitative inversion already reported for FineGym in the main paper ( $r = -0.475$ ), now shown across all 8 datasets with real statistical power rather than being a single-dataset anomaly. UCF-101 and HMDB-51—appearance- and sports-driven domains with large whole-body and camera motion—have the highest flow-derived cutoff frequencies but among the lowest stride-sensitivity, because the action remains identifiable from almost any sampled frame. AUTSL and FineGym show the opposite behavior: their discriminative evidence is localized in handshape changes or brief gymnastics phase events, which are diluted by a whole-frame flow aggregate. We conclude that whole-frame optical-flow frequency content is a systematically *anti-correlated*, not merely noisy, proxy for the temporal demand that matters to a classifier. A class- or region-localized motion signal (e.g., hands in sign language or salient joints in gymnastics) is a natural direction for a more faithful spectral test.

## S13 Additional Robustness Checks

*Observational architecture interpretation.* The architecture comparison in the main paper is intentionally phrased as an empirical association rather than a controlled causal ablation. Attention layout, state-space recurrence, tokenization, pretraining objective, parameter count, and optimization all covary across the evaluated architectures. A definitive mechanism test would require holding the backbone fixed while swapping only temporal aggregation modules. Our contribution is the large controlled sampling sweep showing that, within this architecture pool, models with global divided attention or recurrent state integration are consistently less sensitive to temporal subsampling than CNNs and factorized-attention Transformers.

*Dense-accuracy headroom does not explain TDS.* Because several datasets are evaluated on balanced subsets or fixed 20-clips-per-class protocols, absolute gaps to external leaderboard numbers are not directly comparable. The relevant risk is whether under-optimized model-dataset pairs inflate TDS simply because they have more room to lose accuracy. We therefore compute an internal dense-accuracy headroom: for each dataset, the gap between each model’s stride=1 accuracy and the best stride=1 accuracy achieved by any model in our pool. If under-optimization drove aliasing loss, this headroom should correlate positively

with stride=1→16 drop. It does not: across the 64 model-dataset pairs, Spearman  $\rho = -0.22$  ( $p = 0.079$ ); at the dataset level, mean headroom vs. TDS gives  $\rho = 0.36$  ( $p = 0.385$ ).

**Table S18:** Dense-accuracy headroom check. Headroom is the mean gap from each model’s stride=1 accuracy to the best stride=1 accuracy achieved by any model on the same dataset. It is not significantly correlated with TDS, suggesting that temporal sensitivity is not an artifact of under-optimized dense accuracy.

| Dataset       | Mean stride=1 | Best stride=1 | Mean headroom | TDS  |
|---------------|---------------|---------------|---------------|------|
| FineGym       | 73.8          | 79.9          | 6.1           | 55.9 |
| AUTSL         | 72.9          | 82.3          | 9.3           | 53.0 |
| SSv2          | 42.5          | 52.4          | 9.9           | 27.6 |
| DriveAct      | 67.4          | 73.9          | 6.5           | 21.9 |
| Diving-48     | 40.3          | 53.0          | 12.7          | 19.2 |
| HMDB-51       | 78.2          | 84.0          | 5.8           | 16.6 |
| EPIC-Kitchens | 34.9          | 39.4          | 4.5           | 9.7  |
| UCF-101       | 89.0          | 95.5          | 6.5           | 4.9  |

*Coverage-stride interaction.* Figures in S1 visualize the complete coverage×stride grids. To test additivity, we average accuracy over architectures for representative high-TDS and low-TDS datasets, then decompose the 5×5 mean grid into row/column effects plus an interaction residual. FineGym has a meaningful interaction component (15.0% of grid variance; RMSE 8.3pp): low coverage is poor regardless of stride, while high coverage exposes the stride cliff. UCF-101 is closer to additive (4.0%; RMSE 1.7pp).

**Table S19:** Mean accuracy (%) over architectures for a high-TDS dataset (FineGym) and a low-TDS dataset (UCF-101). Rows are coverage; columns are stride.

| FineGym  |      |      |      |      |      | UCF-101  |      |      |      |      |      |
|----------|------|------|------|------|------|----------|------|------|------|------|------|
| Coverage | s1   | s2   | s4   | s8   | s16  | Coverage | s1   | s2   | s4   | s8   | s16  |
| 10%      | 8.0  | 7.2  | 6.4  | 5.8  | 5.8  | 10%      | 74.5 | 73.2 | 69.5 | 64.2 | 57.7 |
| 25%      | 16.9 | 15.6 | 12.4 | 8.3  | 6.4  | 25%      | 81.7 | 81.6 | 80.1 | 75.0 | 67.8 |
| 50%      | 42.9 | 40.7 | 28.7 | 17.1 | 9.3  | 50%      | 86.6 | 86.6 | 85.7 | 82.9 | 76.6 |
| 75%      | 63.4 | 59.6 | 42.3 | 25.1 | 13.1 | 75%      | 88.4 | 88.7 | 88.5 | 86.8 | 81.4 |
| 100%     | 73.8 | 68.0 | 49.8 | 29.8 | 15.7 | 100%     | 89.0 | 89.0 | 89.0 | 88.1 | 84.1 |

*VideoMamba on AUTSL.* For temporal aliasing, the VideoMamba AUTSL result uses the same coverage×stride protocol as the other temporal results. For the separate spatial cross-resolution table in S9, we omit VideoMamba on AUTSL because the available checkpoint set is not comparable for that specific spatial sweep. This does not drive the VideoMamba robustness conclusion: its mean stride drop is 11.4pp over all 8 datasets and 10.7pp if AUTSL is excluded.

*Entropy-threshold calibration.* The entropy router is not trained; it reuses two fixed inference outputs (4-frame cheap and 16-frame dense) and sweeps a scalar confidence threshold to trace a validation-set Pareto frontier. Thus the reported threshold should not be interpreted as calibrated on an independent test split. The threshold-free baselines (fixed 4f, fixed 8f, fixed 16f) and the oracle bound are included to expose this dependence. In deployment, the threshold should be selected on a held-out calibration split or with nested validation; the current result is best read as evidence that confidence contains useful routing signal, not as a fully validated routing policy.

## References

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